HyLiOSR: Staged Progressive Learning for Joint Open-Set Recognition of Hyperspectral and LiDAR Data

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***Abstract*— The joint classification of hyperspectral images (HSIs) and light detection and ranging (LiDAR) data have seen significant advancements in recent research. However, it would be more practical if we could simultaneously detect the unknown classes in a more realistic open-set scenario. In this article, we introduce a novel open-set recognition (OSR) method for HSI and LiDAR data, termed HyLiOSR, which devises a staged progressive learning strategy to effectively bridge the gap between closed-set and open-set feature distributions within an autoencoder framework. Specifically, for the first stage, the reconstruction-based network is dedicated to accurately modeling each known category by learning multiple Gaussian prototypes, which facilitates OSR by disentangling the distribution of known classes. In the second stage, we actively synthesize samples of unknown classes during the feature extraction phase and create a virtual unknown classifier, enabling the network to effectively differentiate between known and unknown class samples. This approach establishes a distinct separation between known and unknown classes in the latent feature space, thereby enhancing the capability of the frameworks to distinguish between them. Comprehensive experiments conducted on three benchmark datasets demonstrate that the proposed HyLiOSR outperforms existing state-of-the-art methods. The source code will be accessible at https://github.com/B-Xi/TGRS_2025_HyLiOSR.**

***Index Terms*— Autoencoder, Gaussian prototypes, hyperspectral image (HSI), light detection and ranging (LiDAR), open-set recognition (OSR).**

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I. Introduction

With the significant advancements in sensor technology, there has been a continuous evolution of multimodal remote sensing (RS) image datasets. Hyperspectral images (HSIs) have consistently been a favored choice in RS due to their multichannel characteristics, which allow for the effective detection of subtle pixel differences and accurate identification of ground objects [1], [2], [3], [4], [5], [6], [7]. For instance, Gao et al. [8] innovatively proposed an HSI classification method based on depth feature interaction, which effectively extracted discriminative spectral information and exhibited excellent classification performance. However, HSIs are susceptible to external influences, such as cloud and shadow occlusion [9], [10], [11], [12]. In contrast, light detection and ranging (LiDAR) images, with their vertical structure, can effectively capture elevation information in areas affected by cloud or shadow, and they are less sensitive to climatic variations [13]. Consequently, the combined use of HSI and LiDAR has become a crucial area of research and is extensively utilized in various applications within the RS field [14], [15], [16], [17], [18].

Current research is centered on the joint classification of HSI and LiDAR data, which presents considerable advantages. For example, various classes within HSIs may display similar spectral characteristics, resulting in a high degree of similarity at the feature level. However, LiDAR data can provide prior knowledge of the elevation information and geometric texture features of objects, facilitating better differentiation between classes that share similar spectral information. The field has progressed from early machine learning techniques that were limited to extracting shallow, hand-crafted features [19], [20], [21], [22] to more advanced models that utilize deep convolutional neural networks (CNNs) and Transformer architectures [23], [24], [25], which automatically derive robust, high-level features for classification. For instance, Xu et al. [26] developed a two-branch CNN (TB-CNN) to extract the spectral–spatial features from HSI and spatial information

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from LiDAR data, which achieves excellent classification performance. After that, Zhao et al. [27] combined the CNNs with a hierarchical random walk network (HRWN) to improve the classification results. In addition, Hong et al. [28] presented a simple but effective multimodal deep learning baseline using a deep encoder–decoder network architecture (EndNet), which enhances the feature diversity by integrating the HSI-LiDAR information and then enforcing the fused features to reconstruct the multimodal input in turn. Zhang et al. [29] broke through the heterogeneity bottleneck of multisource data, and significantly enhanced the robustness of feature extraction and classification processes through structural optimization transmission. Ding et al. [30] proposed a global–local Transformer network (GLT-Net), which effectively exploits the advantages of the convolution operator in characterizing locally correlated features and the promising capability of Transformer networks in learning long-range dependencies.

Building on the previous discussion, the closed-set joint classification of HSI and LiDAR data has made significant progress and has become increasingly refined. However, in numerous real-world situations, new objects frequently appear that belong to categories not presented in the known training samples. This is especially common in the RS community, where the volume of images and categories is rapidly expanding as more data is gathered. Therefore, it is essential to not only concentrate on the model's accuracy with known classes but also to equip the model with the ability to identify unknown classes, which aligns with the goals of open-set recognition (OSR) [31], [32].

Generally, OSR has garnered significant interest in the traditional domain of computer vision and has found extensive applications in HSI processing [33]. The methodologies in this field can be divided into three main categories: generative, discriminative, and reconstruction-based approaches.

In the realm of generative methods, Zhang et al. [34] applied a domain adaptation method to OSR for the RS image scene. They utilized generative adversarial networks (GANs) [35] to create feature generators and classifiers that learn through adversarial processes. The classifier aims to delineate the boundary between source and target samples, while the feature generator works to distance the target samples from this boundary to identify unknown classes. In addition, Pal et al. [36] integrated meta-learning with GANs to facilitate few-shot OSR in HSI, generating class-conditional adversarial samples for both closed and open spaces under few-shot conditions using two GANs, along with a novel regularizer to enhance performance.

For discriminative methods, Baghbaderani et al. [37] developed a representative-discriminative OSR framework, which projects data from the raw image space into an embedded feature space to improve their representativeness and discriminative capability. Liu et al. [38] introduced an OpenMax layer in deep networks, enabling the estimation of the likelihood that an input belongs to an unknown class while classifying known categories. Subsequently, Pal et al. [39] created a feature extractor with new attention-based residual convolutional blocks to enhance spectral–spatial feature learning from HSI, employing an outlier calibration network to predict anomaly scores, thus eliminating the need for manual threshold settings.

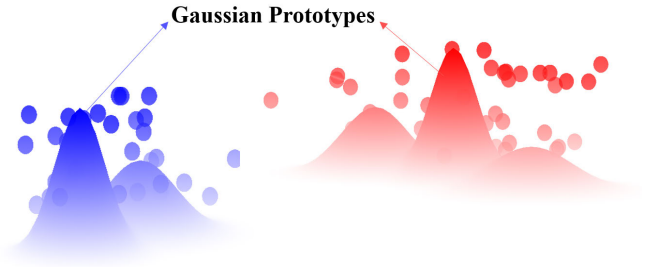


Fig. 1. Schematic visualization of multiple Gaussian prototypes to fit the feature distribution. Different colors represent various classes.

Reconstruction-based methods operate on the premise that autoencoder networks can effectively reconstruct the trained known classes. For unknown classes, the significant reconstruction errors can be leveraged to identify them based on the extreme value theory (EVT) during the inference [40]. For instance, Liu et al. [41] proposed a multitask deep learning approach (MDL4OW) that evaluates the reconstructed data against original samples to detect unknown classes, using a threshold based on EVT during the test for various scenarios. Similarly, Yue et al. [42] introduced a spectral–spatial latent reconstruction (SSLR) framework that simultaneously reconstructs both spectral and spatial features while conducting pixel-wise classification, thereby enhancing feature representations essential for distinguishing between known and unknown classes in HSI data. Following this, Fang et al. [43] also proposed geometric-spectral reconstruction learning (GSRL) for multimodal image OSR tasks, which combines the complementary features of HSI and LiDAR data to navigate the open world more effectively.

Among the approaches discussed, reconstruction-based methods are widely adopted due to their intuitive nature. However, they often fail to adequately represent known-class features within the latent feature space, leading to insufficient representation, even when employing variational autoencoders (VAEs) to model features with specific distributions [44]. This limitation poses significant challenges for the model in distinguishing between known and unknown classes based on reconstruction errors. To address this issue, we propose a method that learns multiple Gaussian prototypes to better capture the complex joint features of HSI and LiDAR for each known class, as illustrated in Fig. 1. Unlike traditional autoencoder frameworks, our approach incorporates a closed-set classifier following the encoder in the initial training stage, which enhances the discrimination of known classes. In the second training stage, we synthesize unknown class samples to decrease the confidence of the closed-set classifier regarding these unknowns and to increase the separation between known and unknown categories. In addition, we establish a virtual unknown classifier to enhance the capability of the model to distinguish between known and unknown samples.

The key contributions of this work are as follows.

1) We introduce a novel staged progressive learning strategy within the autoencoder framework to facilitate joint OSR for HSI and LiDAR data.

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2) In the first learning stage, we employ multiple Gaussian prototypes to represent known categories, effectively modeling the distributions of the intricate integrated HSI and LiDAR latent features and ensuring robust feature representation.
3) The second stage involves fine-tuning training to synthesize unknown classes, followed by a virtual unknown classifier to enhance the model's discriminative capability between known and unknown classes.
4) We performed extensive experiments on three benchmark datasets of paired HSI and LiDAR, demonstrating the superior performance of the proposed HyLiOSR in classifying known classes and detecting unknown samples.

The remainder is organized as follows. Section II reviews the related works. Section III elaborates on our proposed method. Section IV presents the experimental results and analysis. Section V concludes the study.

II. Related Works

The proposed HyLiOSR framework is built upon the reconstruction-based OSR method, with prototype learning integrated into a staged progressive learning strategy. To clarify these concepts, we will discuss both aspects in detail.

A. Reconstruction-Based OSR Method

In OSR tasks, it is commonly assumed that the reconstruction discrepancy can effectively differentiate between known and unknown classes. Because samples from known classes typically exhibit smaller reconstruction errors compared to those from unknown classes that the model has not encountered. Consequently, many OSR methods leverage reconstruction errors within the reconstruction networks to enhance the model's ability to distinguish unknown classes [45]. For instance, Yoshihashi et al. [46] incorporated latent feature representations from the reconstruction network alongside network predictions for classification purposes. They introduced a specialized autoencoder known as the deep reconstruction network, which extracts potential features at each layer. During the training phase, the network is optimized using both classification loss and reconstruction loss, with the classification loss relying on scores derived from the EVT. In the test phase, these classification scores are utilized to identify samples belonging to known and unknown categories.

These approaches laid the groundwork for further advancements in OSR, as demonstrated by Oza and Patel [47], who divided the autoencoder training process into two distinct steps. Initially, they trained the encoder and classifier, linking them to potential features through traditional cross-entropy classification loss. Subsequently, the weights of the encoder and classifier were fixed, and the decoder was trained using a carefully crafted pairwise reconstruction loss. In this phase, the input feature mapping to the decoder is influenced by two elements: features encoded through linear modulation and a condition vector, resulting in a pair of reconstructed images. The decoder was designed to maximize reconstruction errors for label-mismatched pairs while minimizing errors for label-matched pairs. During the test, the coded features of test images were linearly modulated by class-wide condition vectors, allowing for the calculation of reconstruction errors for all known classes, which were then compared against a threshold to determine predicted labels.

Furthermore, Huang et al. [48] combined prototype learning with reconstruction, proposing the reconstruction of class-specific semantic feature maps instead of instance-specific images to enhance the model's semantic interpretability. They developed an autoencoder for each known class in the latent space, which was employed to reconstruct feature maps derived from the backbone encoder of the input image. The pixel reconstruction error mappings specific to each class were treated as logits and transformed into conventional logit vectors. During training, the model was optimized by constraining the classification loss of these logit vectors. In the test, reconstruction errors from the class-specific autoencoder were utilized for recognition. Perera et al. [49] discovered a novel method for leveraging reconstructed information through reconstructed images. They initially trained a generative model (such as an autoencoder) to generate reconstructed images of known classes, subsequently augmenting the original images for classifier training.

B. Prototype Learning

Prototypes generally serve as feature vectors that represent instances or categories within the feature space, which have been widely developed in HSI classification [50], [51], [52], [53], [54]. The traditional prototype design is the K-means algorithm, which identifies a specific instance as the center. In recent years, there has been a growing trend to integrate prototype learning with feature learning using neural network architectures. Prototypes can be considered as learnable parameters that are optimized through custom loss functions in deep learning frameworks. For instance, Wen et al. [55] introduced a center loss to facilitate the learning of prototype centers. Building on this, Yang et al. [56] developed multiple prototype loss functions aimed at learning both representative and discriminative prototypes, which also act as regularization to enhance intraclass compactness. Gu and Li [57] focused on extracting prototypes at various levels of granularity and utilized a locally optimized contrastive learning approach for classification tasks. In our method, we conceptualize each prototype as a Gaussian distribution instead of a simple feature vector. These Gaussian prototypes are learned in a probabilistic manner, integrating both generative and discriminative constraints to enhance the representational power of features in the latent space.

III. Proposed HyLiOSR for Joint OSR

The two-stage training processes of the HyLiOSR are depicted in Figs. 2 and 3, corresponding to the pseudocodes depicted in Algorithms 1 and 2, respectively. In the initial stage, the network is trained to learn multiple Gaussian prototypes for each known class, allowing for a more accurate representation of the complex joint data of HSI and LiDAR. Concurrently, a closed-set classifier is implemented to enhance

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the differentiation of known classes. In the second training phase, we begin by generating simulated samples of unknown classes using the extracted features, effectively creating additional space for the unknown categories. Furthermore, a virtual unknown classifier is introduced to lower the confidence levels assigned by the closed-set classifier to the unknown-class samples, thereby improving the distinction between known and unknown classes. After that, the EVT is employed to classify the samples based on weighted reconstruction errors of the HSI and LiDAR data. The following subsections will detail the two training processes and the reconstruction-based joint HSI-LiDAR OSR configurations.

Algorithm 1 First Learning Stage

Data: HSI data x_1, LiDAR data x_2, number of Gaussian prototypes $\text{Num}_{\text{proto}}$, the dimensionality of the Gaussian prototypes $\text{Dim}_{\text{proto}}$, first-stage training epoch number $\text{Epoch}_{1\text{st}}$

Result: Reconstructed HSI $\widehat{x_1}$ and LiDAR $\widehat{x_2}$

for *epoch* $\leq Epoch_{1st}$ **do**
  $F \leftarrow$ Extract features of batch (x_1, x_2);
  $(\mu_T, \sigma_T, y) \leftarrow$ FC-2, FC-3 and FC-1 (F);
  $(\widehat{x_1}, \widehat{x_2}) \leftarrow$ utilize the decoder in Table I;
  Calculate the cross-entropy and restraint losses;
  Optimizing the losses by gradient descent;
end

A. Multiple Gaussian Prototypes Learning for Known Classes

1) Gaussian Prototype: A Gaussian prototype T is defined as (μ_T, σ_T, y), where y denotes the class label, and μ_T and σ_T represent the mean and variance of the Gaussian distribution associated with that prototype. We employ Gaussian prototypes to describe the distribution of each class within the latent variable space. Specifically, the latent space is structured to include several Gaussian regions corresponding to different classes, each represented by multiple Gaussian prototypes. In contrast to earlier conditional VAE-based methods [40], [46], which typically assign a single Gaussian prototype to each class, our approach utilizes multiple Gaussian prototypes to better accommodate the intricate data distributions present in HSI and LiDAR scenarios. This implies a one-to-many relationship between classes and Gaussian prototypes, allowing a single class to be represented by several Gaussian prototypes due to its complex distribution. Consequently, for a dataset containing c class, we assign n Gaussian prototypes to each class, leading to a total of $n \times c$ prototypes, which will be elaborated upon in the parameter settings section.

2) Restraint Loss: To guarantee the discrimination of the extracted features, we adopt a fully connected (FC) layer (seen as FC-1 in Fig. 2) and back-propagate the cross-entropy loss. Particularly, to ensure that the multiple Gaussian prototypes accurately represent their respective classes, we develop restraint loss functions to enforce the necessary constraints. By applying Bayesian inference to approximate $p_\theta(x|y)$ where

TABLE I
ARCHITECTURE OF THE ENCODER AND DECODER LAYERS

Encoder				
Layer_name	**Input shape**	**Kernel size**	**Padding**	**Output shape**
Conv2d-1	(-1,63, 9, 9)	(3,3,32)	Valid	(-1, 32, 7, 7)
Conv2d-2	(-1,2, 9, 9)	(3,3,32)	Valid	(-1, 32, 7, 7)
Concat	(32,7,7)×2	-	-	(32+32,7,7)
Conv2d-3	(-1, 64, 7, 7)	(3,3,64)	Same	(-1, 64, 7, 7)
Conv2d-4	(-1, 64, 7, 7)	(3,3,64)	Same	(-1, 64, 7, 7)
Conv2d-5	(-1, 64, 7, 7)	(3,3,64)	Same	(-1, 64, 7, 7)
AvgPool	(-1, 64, 7, 7)	-	-	(-1, 64, 1, 1)
Flatten	(-1, 64, 1, 1)	-	-	(-1, 64)
FC-1(Known)	(-1, 64)	-	-	(-1, 6)
FC-2(μ)	(-1, 64)	-	-	(-1, $\text{Dim}_{\text{proto}}$)
FC-3($\log(\sigma^2)$)	(-1, 64)	-	-	(-1, $\text{Dim}_{\text{proto}}$)
FC-4(Unknown)	(-1, 64)	-	-	(-1, $\text{Num}_{\text{pseudo}}$)
Decoder				
Layer_name	**Input shape**	**Kernel size**	**Padding**	**Output shape**
Reshape-1	(-1, 64)	-	-	(-1, 64,1,1)
Deconv2d-1	(-1, 64,1,1)	(1,1,64)	Valid	(-1, 64,1,1)
Deconv2d-2	(-1, 64,1,1)	(3,3,64)	Valid	(-1, 64,3,3)
Deconv2d-3	(-1, 64,3,3)	(3,3,64)	Valid	(-1, 64,5,5)
Deconv2d-4	(-1, 64,5,5)	(3,3,64)	Valid	(-1, 64,7,7)
Split	(32+32,7,7)	-	-	(32,7,7)×2
Deconv2d-5	(-1,32,7,7)	(3,3,63)	Valid	(-1,63,9,9)
Deconv2d-6	(-1,32,7,7)	(3,3,1)	Valid	(-1,2,9,9)

x denotes the original data, we derive its evidence lower bound (ELBO) as follows:

$$\begin{aligned} L_{\text{ELBO}} &= \mathbb{E}_{q_\phi(z,T|x,y)}\left[\log \frac{p_\theta(x,z,T|y)}{q_\phi(z,T|x,y)}\right] \\ &= \mathbb{E}_{q_\phi(z|x)q(T|z,y)}\left[\log \frac{p_\theta(x|z)p(z|T)p(T|y)}{q_\phi(z|x)q(T|z,y)}\right] \\ &= \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] \\ &\quad - \mathbb{E}_{q(T|z,y)}[\text{KL}(q_\phi(z|x)\|p(z|T))] \\ &\quad - \mathbb{E}_{q_\phi(z|x)}[\text{KL}(q(T|z,y)\|p(T|y))] \end{aligned} \tag{1}$$

where z represents the latent variable following the Gaussian distribution. The first term, $\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$ serves as an expectation in variational inference, measuring the fit between the reconstructed data $f(z;\theta)$ and the original data x under the variational distribution $q_\phi(z|x)$. For simplicity, we directly utilize the error between the original and reconstructed data to define the reconstruction loss

$$\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] = \|\widehat{x_1} - x_1\|_2 + \epsilon \cdot \|\widehat{x_2} - x_2\|_2 \tag{2}$$

where $\widehat{x_1}$ and $\widehat{x_2}$ represent the reconstructed HSI and LiDAR data, respectively. ϵ is a weighting hyperparameter. The second term is designed to ensure that the latent feature z is close to the Gaussian prototypes of the corresponding categories, which can be expressed as

$$\begin{aligned} &\mathbb{E}_{q(T|z,y)}\left[\text{KL}\left(q_\phi(z|x)\|p(z|T)\right)\right] \\ &\quad = \mathbb{E}_{q(T|z,y)}[\text{KL}(\mathcal{N}(z;\mu_x,\sigma_x)\|\mathcal{N}(z;\mu_T,\sigma_T))] \end{aligned} \tag{3}$$

where Kullback–Leibler (KL) divergence is adopted to constrain the latent variable z to align with its Gaussian prototypes. μ_x and σ_x are obtained by two individual FC layers (seen as FC-2 and FC-3 in Fig. 2). This optimization objective encourages z not only to approach the Gaussian prototype of its corresponding category but also to cluster around it, forming a cohesive group.

The third component aims to balance the learning process across multiple prototypes for each category, which can be

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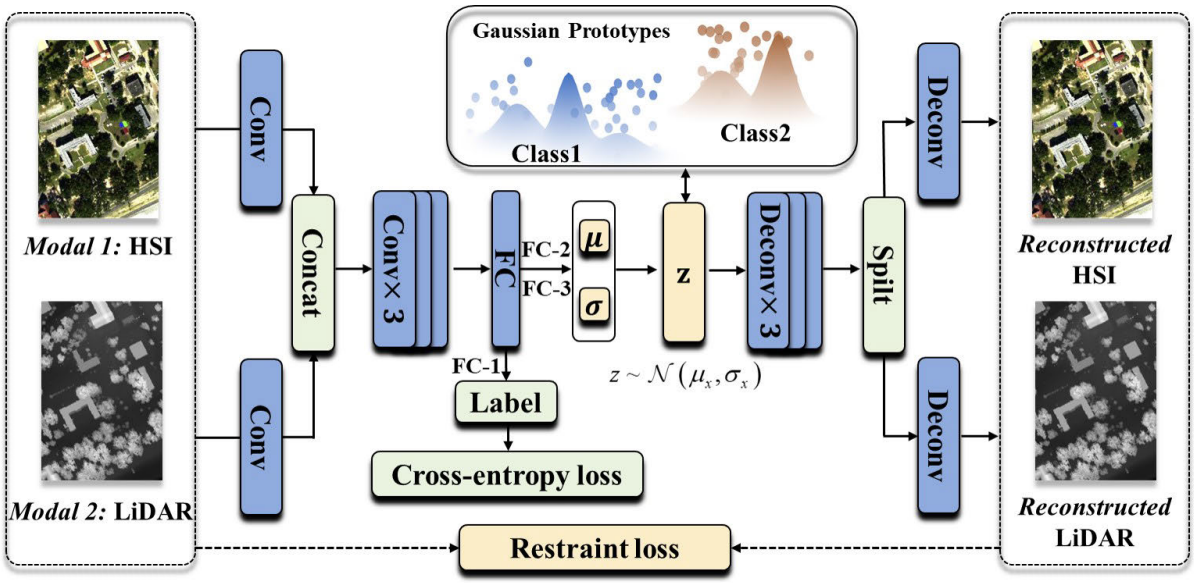


Fig. 2. HSI and LiDAR joint OSR networks in the first learning stage.

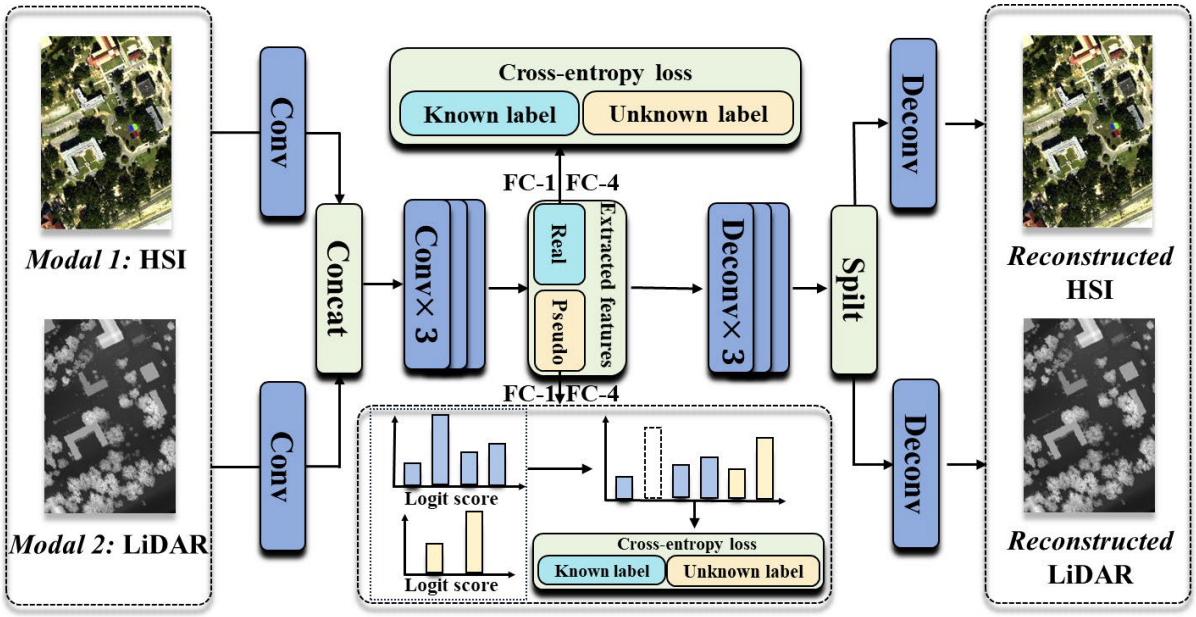


Fig. 3. HSI and LiDAR joint OSR networks in the second learning stage.

expanded as follows:

$$\begin{aligned}&\mathbb{E}_{q_\phi(z|x)}[\mathrm{KL}(q(T|z,y)\|p(T|y))]\\&=\mathbb{E}_{q_\phi(z|x)}\sum_{T\in T_y}\left[-q\left(T\middle|z,y\right)\log\frac{q(T|z,y)}{p(T|y)}\right]\\&=\mathbb{E}_{q_\phi(z|x)}\sum_{T\in T_y}[-q(T|z,y)\log K(q(T|z,y))]\end{aligned}\tag{4}$$

where K represents the number of known classes, and T_y denotes the prototypes for category y. The term $p(T|y) = 1/K$ indicates that the selection probability for each prototype T of class y is uniform, without bias toward any specific prototype. In addition, $q(T|z, y)$ is the conditional probability of selecting a particular prototype T given the latent variable z and category y. This formulation primarily ensures that the probabilities of prototypes for each class are uniformly distributed, enabling the model to consider all prototypes equally during the learning process. This approach enhances the model's ability to capture data diversity and prevents it from overly relying on a limited set of prototypes when representing classes.

B. Space Preservation for Unknown Classes

1) Synthesis of Unknown Classes: To enhance the capability of the frameworks to manage unknown classes, we implement *Beta* distribution-based sampling perturbations at the end of the encoder's convolutional layers. Specifically, the *Beta* distribution [58] is a probability distribution defined

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Algorithm 2 Second Learning Stage

Data: HSI data x_1, LiDAR data x_2, second-stage training epoch number $\text{Epoch}_{2\text{nd}}$
Result: Reconstructed HSI $\widehat{x_1}$ and LiDAR $\widehat{x_2}$

```
Frozen the model wights of Gaussian prototypes;
for epoch ≤ Epoch_2nd do
    F_1 ← Extract features of 1st-half batch (x_1, x_2) ;
    Label_pre ← FC-1, FC-4(F_1) ;
    L_1 ← Cross-entropy(Label_pre, gt) ;
    F_2 ← Extract features of 2nd-half batch (x_1, x_2) ;
    F_pseudo ← Utilizing formula (7)(F_2) ;
    Label_pre ← FC-1, FC-4(F_pseudo) ;
    L_2 ← Cross-entropy(Label_pre, Pseudo label) ;
    Optimizing the losses by gradient descent;
end
```

over a continuous variable $x \in [0, 1]$, characterized by two parameters, a and b, and is expressed as

$$\begin{aligned}\text{Beta}(\mu|a,b) &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\mu^{a-1}(1-\mu)^{b-1} \\ &= \frac{1}{B(a,b)}\mu^{a-1}(1-\mu)^{b-1} \end{aligned} \tag{5}$$

where $\Gamma(x)$ denotes the gamma function, defined as $\Gamma(x) = \int_0^{+\infty} t^{x-1}e^{-t}dt$, and $B(a, b)$ is given by

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}. \tag{6}$$

In specific, we keep the first-half data of each batch flows as the first learning stage and synthesize new features using the second-half ones by the formula

$$\tilde{\mathbf{x}}_{\text{new}} = \lambda\phi_{\text{pre}}(\mathbf{x}_i) + (1-\lambda)\phi_{\text{pre}}(\mathbf{x}_j), \quad y_i \neq y_j \tag{7}$$

where λ is sampled from the *Beta* distribution. This random sampling process results in perturbed features that can be interpreted as deep features of the unknown classes. The rationale and feasibility of this approach are discussed in the ablation experimental section.

2) Classifier for Unknown Classes: In the closed-set classification tasks, the logit score indicating the predicted probability for each category is derived from the FC-1 layer and subsequently normalized using the softmax function. Although the network trained in the first stage effectively fits known classes, it still struggles to distinguish unknown classes. To address this, we develop an extra unknown class classifier with a new FC layer (seen as FC-4 in Fig. 3) specifically designed to predict the probability of unknown classes.

For the known class samples, we compute the cross-entropy loss as the initial learning phase. In contrast, for the pseudo-unknown class samples generated, as depicted in Fig. 4, we first set the highest logit score from the closed-set classifier to zero. Then, we concatenate the logit scores produced by the two classifiers and normalize the combined logit scores using the softmax function to calculate the cross-entropy loss. In this case, the pseudo groundtruth label corresponds to the unknown category with the highest score from the unknown class classifier. This approach enables the combined classifier to identify new class samples effectively, preventing it from assigning high logit scores to known categories when faced with unknown class samples during the test.

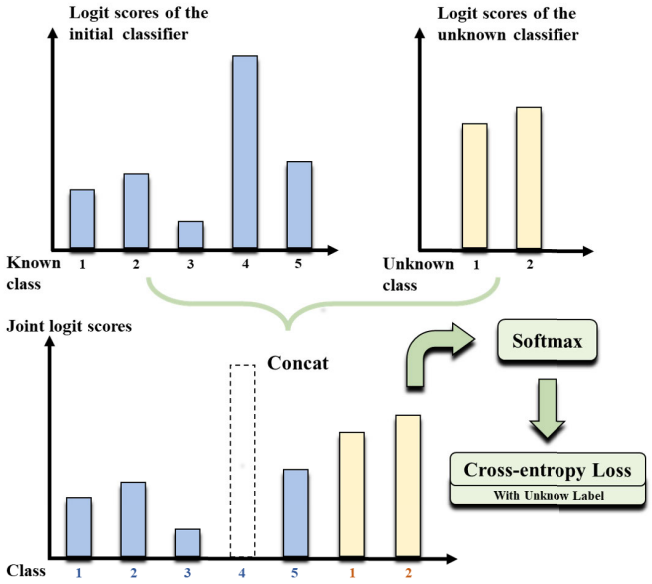


Fig. 4. Illustration of the combined classifier for the unknown class samples.

C. Reconstruction-Based Joint OSR Setups

1) Open-Set Settings and Network Architectures: Consider a training dataset consisting of labeled samples from K distinct known classes, represented as $D_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{n}$. Each sample pair (x_i, y_i) includes an input $x_i \in \mathcal{X}$ and its corresponding label $y_i \in \mathcal{K} = 0, \ldots, K-1$. In scenarios involving open-world or open-set conditions, test data can be drawn not only from the known classes K but also from a set of unknown classes $\mathcal{U} = \{K, \ldots, K+U-1\}$, where U denotes the number of unknown classes. The primary challenge in OSR is to differentiate between known and unknown classes. Specifically, when presented with an unlabeled test sample x, the model first determines if x belongs to the known classes K. If it is classified as a known class, the model outputs the specific class label; otherwise, it identifies x as belonging to the unknown class.

In the proposed frameworks, the distributions of known classes are explicitly modeled by the autoencoder during the training phase. This allows for the differentiation of unknown class samples from known class samples based on the reconstruction error because known class samples typically exhibit lower reconstruction errors than unknown class samples. If a sample does not conform to any of the modeled known class distributions, it can be rejected as an unknown class. Conversely, known class samples can be classified according to their respective distributions. The architecture of the encoder and decoder layers is detailed in Table I, where the FC-1, FC-2, FC-3, and FC-4 can be referred to Figs. 2 and 3. $\text{Dim}_{\text{proto}}$ indicates the dimensionality of the prototypes.

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TABLE II
LABELED SAMPLES INFORMATION FOR THE TR DATASET

No.	Class name	Num
1	Apple Trees	4034
2	Buildings	2903
3	Ground	479
4	Wood	9123
5	Vineyard	10501
6	Roads	3174
Unknown1	Grass	1891
Unknown2	Soil	1024
-	**Total**	**33129**

2) Extreme Value Theory: To identify unknown instances, the reconstruction error is modeled using EVT, which typically employs the Weibull distribution [59] to fit long-tailed error data. In our experiments, the number of tails σ is set to 20 as in research [41] across all datasets. This distribution allows us to classify instances with reconstruction errors that fall into the tail regions as unknown classes. In specific, the cumulative probability density function of the Weibull distribution is given by

$$\mathbf{W}^{\text{cdf}}_{\alpha,\beta}(z) = \begin{cases} 1 - e^{-(z/\beta)^{\alpha}}, & (z \geq 0) \\ 0, & (z < 0) \end{cases} \tag{8}$$

where α and β are the shape and scale parameters, and z refers to the loss value corresponding to a pixel. We utilize the mean average error (MAE) loss to compute the reconstruction errors for the HSI and LiDAR data, denoted as L_{HSI} and L_{LiDAR}, respectively. The LiDAR data, which contains elevation information, is generally less precise than the spectral features captured by the HSI. Therefore, we model the overall error of the joint HSI and LiDAR features as $L_{\text{recon}} = L_{\text{HSI}} + \epsilon \cdot L_{\text{LiDAR}}$ and set $\epsilon < 1$.

In addition, a threshold γ is established here to differentiate between known and unknown classes

$$y_{\text{pred}} = \begin{cases} y_{\text{pred}}, & \mathbf{W}^{\text{cdf}}_{\alpha,\beta}(z) < \gamma \\ \text{unknown}, & \text{else}. \end{cases} \tag{9}$$

Following [41], we set γ to 0.5 in this study.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets

To assess the effectiveness of our approach, we conducted experiments using three publicly available HSI-LiDAR paired datasets: Trento, Houston University 2013, and MUUFL. Detailed information about these datasets is presented in Tables II–IV, with visual representations in Figs. 5–7, respectively. Categories marked with an asterisk (*) are designated as unknown categories within the dataset.

1) Trento Dataset (TR): This dataset was captured in a rural area of southern Trento, Italy, and includes both HSI and LiDAR data with dimensions of 166 × 600 pixels and a spatial resolution of 1 m. The HSI was collected using the AISA Eagle sensor, which operates across 63 spectral bands ranging from 0.42 to 0.99 μm. The LiDAR data was obtained using the Optech ALTM 3100EA sensor, designed for airborne laser topographic mapping. It contains a total of 30 214 groundtruth samples classified into six known categories, with "Grass" and "Soil" added as unknown categories, representing

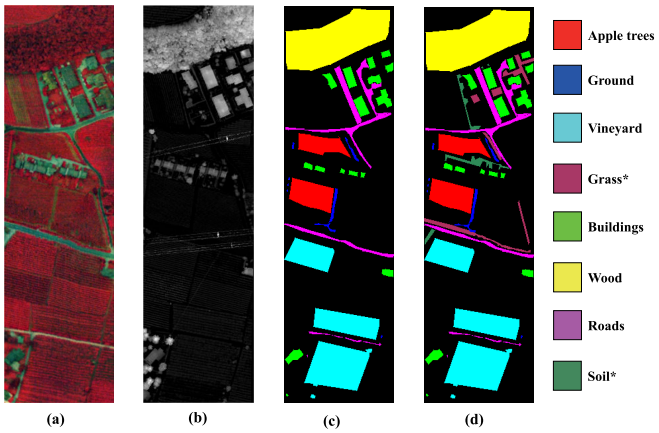


Fig. 5. TR dataset and its label visualization. (a) False-color image of HSI. (b) LiDAR image. (c) Groundtruth without unknown classes. (d) Groundtruth with unknown classes.

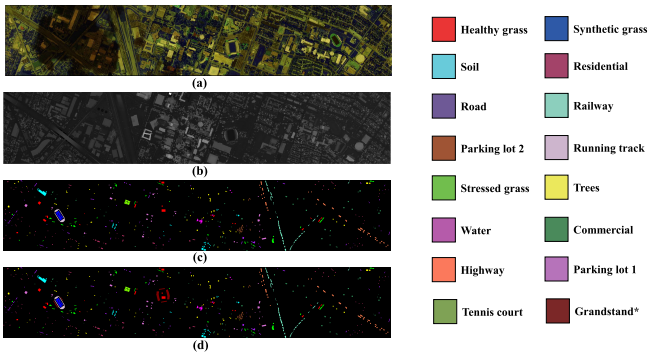


Fig. 6. HU dataset and its label visualization. (a) False-color image of HSI. (b) LiDAR image. (c) Groundtruth without unknown classes. (d) Groundtruth with unknown classes.

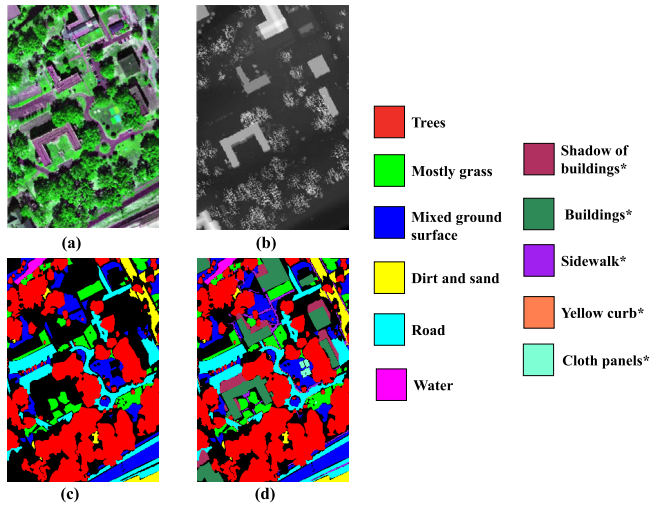


Fig. 7. MU dataset and its label visualization. (a) False-color image of HSI. (b) LiDAR image. (c) Groundtruth without unknown classes. (d) Groundtruth with unknown classes.

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TABLE III
LABELED SAMPLES INFORMATION FOR THE HU DATASET

No.	Class name	Num
1	Healthy Grass	1251
2	Stressed Grass	1254
3	Synthetic Grass	697
4	Trees	1244
5	Soil	1242
6	Water	325
7	Residential	1268
8	Commercial	1244
9	Road	1252
10	Highway	1227
11	Railway	1235
12	Parking Lot 1	1233
13	Parking Lot 2	469
14	Tennis Court	428
15	Running Track	660
Unknown	Grandstand	1519
-	**Total**	**16548**

TABLE IV
LABELED SAMPLES INFORMATION FOR THE MU DATASET

No.	Class name	Num
1	Trees	23246
2	Mostly grass	4270
3	Mixed ground surface	6882
4	Dirt and sand	1826
5	Road	6687
6	Water	466
Unknown1	Buildings	2233
Unknown2	Shadow of buildings	6240
Unknown3	Sidewalk	1385
Unknown4	Yellow curb	183
Unknown5	Cloth panels	269
-	**Total**	**53687**

8.80% of the total samples. The detailed information is shown in Table II.

2) Houston University 2013 Dataset (HU): Compiled by the National Center for Airborne Laser Mapping (NCALM) in June 2012, this dataset features imagery captured over the University of Houston and its urban surroundings using the Compact Airborne Spectrographic Imager (CASI) sensor. Released as part of the IEEE Geoscience and Remote Sensing Society's 2013 Data Fusion Competition, it includes HSI and a LiDAR-derived Digital Surface Model (DSM), both measuring 349 × 1905 pixels with a spatial resolution of 2.5 m. The HSI consists of 144 spectral bands covering wavelengths from 0.38 to 1.05 μm, totaling 15 029 groundtruth samples in 15 classes. To conduct the open-set classification experiments, the "Grandstand" category is included as an unknown category, accounting for 9.17% of the total categories. The detailed information is depicted in Table III.

3) MUUFL Dataset (MU): Collected in November 2010 at the University of Southern Mississippi's Gulfport campus, this dataset includes HSI and a DSM derived from LiDAR, both with dimensions of 325 × 220 pixels. The HSI was captured using the ITRES Compact Airborne Spectral Imager (CASI-1500) sensor, featuring 64 spectral channels from 0.38 to 1.05 μm and a spatial resolution of 0.54 × 1.0 m. The LiDAR data was gathered using an ALTM sensor operating at a 1064 nm laser wavelength, achieving a spatial resolution of 0.60 × 0.78 m. This dataset comprises 11 labeled classes and a total of 53 687 samples, with categories such as "Building shadow," "Building," "Sidewalk," "Yellow curb," and "Cloth panels" classified as unknown, making up 19.20% of the total samples. The detailed information is listed in Table IV.

B. Experiment Setup

To evaluate the effectiveness of our proposed method, we conducted a comparison with seven leading techniques, which include four HSI-LiDAR closed-set classifiers: TB-CNN [26], HRWN [27], EndNet [28], and GLT-Net [30], as well as three OSR methods: CROSR [46], MDL4OW [41], and GSRL [43]. For the closed-set methods, we focused on the outputs from their final softmax layer, designating samples with a maximum probability value below 0.5 as belonging to the unknown category. Meanwhile, for the OSR methods, we input the HSI and LiDAR data into the network for feature extraction and applied the same fusion technique as utilized in our approach.

During the training process, we monitored the changes in loss values, leading us to set the number of epochs for the first learning stage to 170, 1100, and 1100 for the three datasets, while the second stage epochs were set to 130, 1000, and 1000. The batch size for the first stage was consistently set to 32, whereas for the second stage, it was adjusted to 500, 200, and 400 across the three datasets. The learning rates were established at 1.0 for the first stage and 0.1 for the second stage. The multiple Gaussian prototypes encoded in the hidden variables were configured with counts of 6, 4, and 1, and their respective dimensions were set to 45, 5, and 15. For the unknown classifier in the second stage, we designated 3, 2, and 2 categories. In addition, the coefficients ϵ for the reconstruction loss were set to 0.05, 0.1, and 0.1 for the three datasets. The patch size was maintained at 9 × 9, consistent with [43]. Given the limited sample size conditions in RS, we selected 20 samples per class for training, with the remaining samples utilized for testing across all three datasets.

C. Experiments Results Analysis

1) Experiments on Dataset TR: Table V displays the experimental results for various comparison algorithms applied to the TR dataset. The findings indicate that the TR dataset demonstrates the best performance among the three datasets, attributed to its compact data distributions. Notably, the TB-CNN, HRWN, EndNet, and GLT-Net methods that were not specifically trained for OSR show limited ability to recognize unknown classes. Among these, the EndNet and GLT-Net achieve higher OAs, despite their low performance in recognizing unknown classes, due to their strong closed-set classification capabilities. The CROSR, an earlier classical method designed for OSR, exhibits limited performance as it was not initially intended for RS images. On the other hand, the MDL4OW demonstrates improved results. Both the GSRL and our proposed HyLiOSR, which are tailored for

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TABLE V
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON TR DATASET

Class	TB-CNN [26]	HRWN [27]	EndNet [28]	GLT-Net [30]	CROSR [46]	MDL4OW [41]	GSRL [43]	HyLiOSR
1	0.00	0.00	66.21	98.93	64.25	87.95	99.13	**99.87**
2	96.36	96.07	93.87	95.80	95.76	93.52	95.42	**96.56**
3	87.47	89.77	93.95	92.91	**97.29**	91.91	94.37	97.12
4	99.94	**100.00**	99.89	96.00	99.67	92.97	**100.00**	**100.00**
5	99.12	91.83	99.97	100.00	68.72	**100.00**	**100.00**	99.93
6	83.01	84.59	85.85	**90.67**	76.37	67.12	79.11	72.77
Unknown	6.37	8.82	1.37	3.05	27.14	46.33	53.31	**57.94**
OA	77.53 ±3.98	75.05 ±2.86	85.19 ±4.21	89.13 ±1.13	76.56 ±1.56	83.71 ±3.12	91.23 ±2.51	**93.41 ±0.54**
AA	68.93 ±2.11	68.38 ±2.77	77.30 ±3.83	83.91 ±0.98	75.60 ±0.82	78.42 ±1.78	88.48 ±1.66	**89.33 ±0.89**
Kappa	70.36 ±3.44	69.73 ±3.02	80.92 ±3.91	86.39 ±1.03	70.72 ±1.87	81.52 ±1.36	90.52 ±1.83	**91.54 ±0.62**

TABLE VI
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON HU DATASET

Class	TB-CNN [26]	HRWN [27]	EndNet [28]	GLT-Net [30]	CROSR [46]	MDL4OW [41]	GSRL [43]	HyLiOSR
1	79.94	88.09	89.92	95.36	93.92	**95.04**	90.11	92.49
2	74.96	84.85	**97.69**	93.54	62.44	85.25	96.73	97.64
3	99.13	**100.00**	99.57	**100.00**	96.56	99.71	99.86	**100.00**
4	97.75	91.88	97.59	94.77	98.47	97.43	98.7	**98.82**
5	49.28	99.28	**100.00**	99.11	68.68	99.76	**100.00**	98.87
6	**99.08**	90.46	84.92	89.85	68.30	95.38	98.69	96.62
7	77.29	85.09	91.55	**92.59**	79.81	83.83	90.73	90.96
8	41.80	67.77	89.39	**90.59**	34.81	64.14	64.63	74.52
9	46.57	71.57	86.02	82.83	16.21	78.83	**94.01**	87.68
10	26.81	16.71	72.13	83.62	11.90	51.04	83.22	**84.79**
11	14.90	24.70	93.60	**97.17**	3.24	84.21	84.21	90.90
12	31.95	5.92	87.67	84.83	2.14	31.30	**92.13**	87.90
13	60.13	89.77	91.47	**95.31**	87.85	82.94	93.40	72.71
14	82.00	98.13	92.76	97.66	69.16	97.20	93.25	**98.64**
15	99.55	96.06	99.70	**100.00**	99.39	99.24	99.55	96.97
Unknown	3.03	1.05	6.12	2.31	13.80	18.44	25.31	**54.05**
OA	55.03 ±1.87	63.00 ±2.19	83.10 ±2.19	83.82 ±1.31	50.57 ±1.01	81.32 ±0.78	84.84 ±0.34	**87.57 ±1.69**
AA	61.51 ±2.15	69.77 ±1.98	85.85 ±1.98	87.89 ±1.98	56.66 ±0.78	84.99 ±0.56	83.97 ±0.48	**88.93 ±1.40**
Kappa	51.95 ±1.97	60.47 ±1.88	81.85 ±1.88	83.70 ±1.88	47.40 ±0.52	79.71 ±0.67	**87.68 ±0.62**	86.63 ±1.82

HSI and LiDAR joint OSR, achieve superior metrics in OA and unknown class accuracy compared to other methods, with our approach being 2.18% more accurate in OA and 4.63% more accurate in unknown class recognition. In addition, our method shows less variability across multiple experiments, underscoring its validity and robustness.

2) Experiments on Dataset HU: Table VI outlines the experimental results for various comparison algorithms on the HU dataset. Compared to the TR and MU datasets, the HU dataset presents greater challenges due to a higher number of known categories and the presence of noise in the HSI. Nevertheless, our method demonstrates superior metrics and robustness compared to other approaches. Similar to the TR dataset, the TB-CNN, HRWN, EndNet, and GLT-Net exhibit relatively low accuracy in recognizing unknown classes. Furthermore, these methods show significant discrepancies in closed-set classification due to the sparse distribution of the samples. The GLT-Net performs reasonably well, achieving an OA of 83.82%, albeit with limited unknown class recognition. In addition, CROSR, MDL4OW, and GSRL also display relatively low unknown class classification accuracy, with GSRL performing slightly better at 25.31% accuracy, which is still insufficient for effectively addressing the challenges of open-world classification. In contrast, our method achieves the highest accuracy in recognizing unknown classes and attains optimal OA. The unknown class accuracy surpasses that of the second-best GSRL by 30% approximately, demonstrating our method's capability to manage complex recognition scenarios involving a large number of closed-set classes. Notably, it also excels in known class classification accuracy.

3) Experiments on Dataset MU: The experimental results for various algorithms applied to the MU dataset are summarized in Table VII. In contrast to the other two datasets, the MU dataset features more unknown categories alongside significant imbalances within the known categories. This complexity poses a challenge for the model in distinguishing between known and unknown classes. From Table VII, it is evident that the primary difficulty in classifying known classes arises with the "Mixed ground surface" category, where the performance of all methods is notably limited. This challenge stems from the introduction of open-set unknown classes, making "Mixed ground surface" easily confused with unknown categories. Among the compared methods, CROSR demonstrates the best handling of this issue, likely benefiting from its classical

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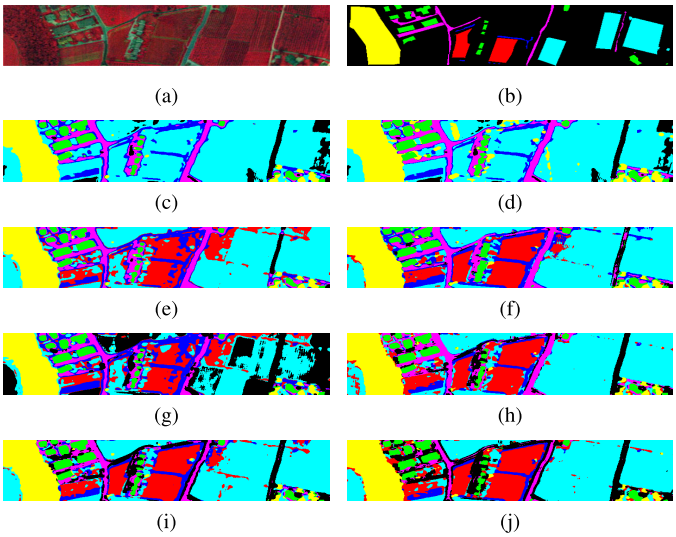


Fig. 8. Classification maps of the TR dataset (black represents unknown classes). (a) False-color image. (b) Groundtruth map. (c) TB-CNN. (d) HRWN. (e) EndNet. (f) GLT-Net. (g) CROSR. (h) MDL4OW. (i) GSRL. (j) HyLiOSR.

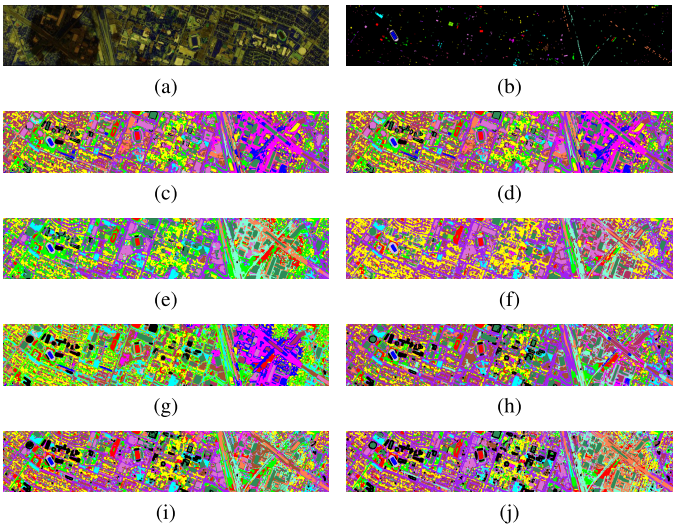


Fig. 9. Classification maps of the HU dataset (black represents unknown classes). (a) False-color image. (b) Groundtruth map. (c) TB-CNN. (d) HRWN. (e) EndNet. (f) GLT-Net. (g) CROSR. (h) MDL4OW. (i) GSRL. (j) HyLiOSR.

unknown class discrimination model. Notably, the proposed method achieves the highest metrics for OA, AA, and Kappa, with about a 20% increase in unknown class accuracy compared to the second-best method.

For visual comparison, we present the best classification maps from various methods in the three datasets in Figs. 8–10, which align with the results shown in Tables V–VII. The groundtruth is also included for easier comparison. In the TR dataset classification maps, the main classification challenges are observed in the "Synthetic grass" and unknown categories, as their spatial proximity leads many methods to misclassify the unknown category as the known class. In addition, "Apple trees" and "Vineyard" are often confused, such as TB-CNN and HRWN completely misidentifying "Apple trees" as "Vineyard," resulting in a classification accuracy of 0 for that class. Furthermore, while CROSR can effectively identify known classes as unknown, its accuracy for unknown classes remains low, highlighting a common risk in OSR tasks, which adversely affects the accuracy of known classes. In the HU dataset classification maps, the primary classification difficulties are found in the categories of "Road," "Stressed grass," and "Trees," with many methods struggling to differentiate among these categories. The HU dataset's larger spatial size and complex categories further exacerbate the challenges and risks of recognizing unknown categories. In the MU dataset, the classification difficulties are particularly evident in the "Mixed ground surface" and "Road" categories, where the TB-CNN method shows the poorest performance. In addition, we observe that many methods struggle to accurately identify unknown categories due to an overly high confidence level in known categories, leading to misclassifications of known classes as unknown, which is a prevalent challenge in the OSR tasks.

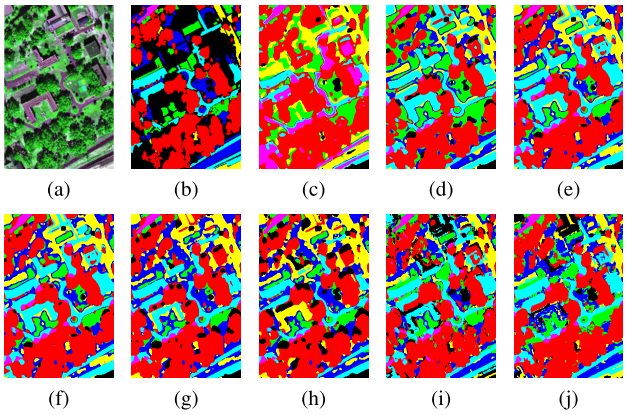


Fig. 10. Classification maps of the MU dataset (black represents unknown classes). (a) False-color image. (b) Groundtruth map. (c) TB-CNN. (d) HRWN. (e) EndNet. (f) GLT-Net. (g) CROSR. (h) MDL4OW. (i) GSRL. (j) HyLiOSR.

D. Ablation Experiments

In our proposed staged progressive learning strategy, the use of multiple Gaussian prototypes is intended to align more accurately with the distribution of known categories. At the same time, the unknown space preservation module enhances the differentiation between known and unknown classes. This section presents ablation experiments to validate the effectiveness of these two learning stages. Specifically, we refer to the multiple Gaussian prototypes generation module as B and the unknown space preservation module as C, whereas the reconstruction framework that excludes these two modules is labeled as A. Consequently, the complete structure of our proposed approach is represented as ABC. We assess the contributions of the two modules by systematically removing modules C and B, with the comparative experimental results detailed in Tables VIII–X. The results indicate that both modules significantly enhance classification and recognition accuracy in open-set scenarios. The absence of these modules results in a decline in the accuracy of unknown class recognition, as well as in OA, AA, and Kappa, and even the stability of performance.

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TABLE VII
CLASSIFICATION PERFORMANCE (%) OF VARIOUS METHODS ON MU DATASET

Class	TB-CNN [26]	HRWN [27]	EndNet [28]	GLT-Net [30]	CROSR [46]	MDL4OW [41]	GSRL [43]	HyLiOSR
1	90.69	87.04	92.01	**92.31**	84.59	82.41	86.28	87.85
2	84.87	**90.35**	76.58	87.52	59.90	71.78	82.04	77.42
3	1.05	41.76	58.81	63.18	**78.87**	75.33	65.19	66.29
4	66.21	69.44	79.46	91.35	78.31	**93.82**	72.07	75.70
5	56.20	75.91	88.33	93.03	71.49	73.81	85.94	**88.46**
6	95.28	85.62	**98.93**	98.28	98.71	98.71	96.14	96.74
Unknown	3.37	7.55	2.31	3.27	11.41	21.11	25.41	**45.20**
OA	56.88 ±2.71	64.42 ±3.98	68.32 ±2.91	71.21 ±0.99	58.31 ±1.64	67.66 ±3.11	71.21 ±2.87	**75.81 ±0.90**
AA	56.81 ±2.94	65.52 ±3.51	69.87 ±3.12	75.56 ±1.03	59.91 ±1.42	71.00 ±2.93	72.87 ±2.56	**76.81 ±2.02**
Kappa	41.75 ±2.86	52.87 ±3.18	66.56 ±2.78	61.57 ±1.01	51.31 ±1.32	57.38 ±2.87	63.44 ±1.32	**67.79 ±1.20**

TABLE VIII
ABLATION EXPERIMENTAL RESULTS ON THE TR DATASET

Class	A	AB	ABC
Unknown	52.68	57.48	**59.06**
OA	92.83 ±0.42	93.25 ±0.68	**93.40 ±0.54**
AA	88.57 ±0.77	89.12 ±1.13	**89.33 ±0.89**
Kappa	90.84 ±0.54	91.37 ± 0.88	**91.57 ±0.69**

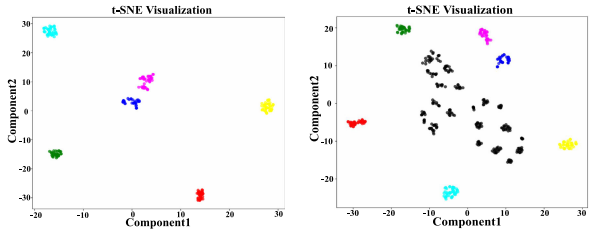
TABLE IX
ABLATION EXPERIMENTAL RESULTS ON THE HU DATASET

Class	A	AB	ABC
Unknown	32.51	50.97	**54.05**
OA	85.40 ±1.89	85.01 ±2.20	**87.57 ±1.69**
AA	87.73 ±1.26	86.55 ±1.88	**88.93 ±1.40**
Kappa	84.32 ±2.03	83.88 ±2.37	**86.63 ±1.82**

TABLE X
ABLATION EXPERIMENTAL RESULTS ON THE MU DATASET

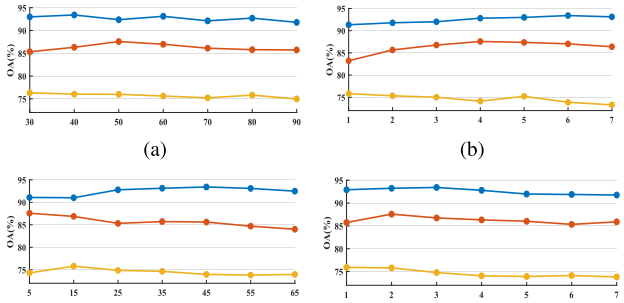
Class	A	AB	ABC
Unknown	36.66	44.55	**45.20**
OA	73.57 ±1.53	75.01 ±1.12	**75.81 ±0.90**
AA	74.56 ±1.57	76.01 ±2.34	**76.83 ±2.02**
Kappa	64.60 ±2.01	66.78 ±1.57	**67.79 ±1.20**

To further illustrate the validity of the synthesized unknown classes, we employ the t-SNE method to visualize the high-dimensional features generated before the FC layer, allowing us to analyze their distribution. Taking the TR dataset as an example, Fig. 11 displays the feature distributions before and after the unknown sample synthesis. From Fig. 11(a), it is observed that the closed-set classifier establishes a clear categorization interval, and the interclass features are disentangled following the first learning stage. Our experimental findings also reveal that the OA for closed-set categories is relatively high, confirming the effectiveness of our approach. In addition, the synthesized unknown samples, depicted in black in Fig. 11(b), effectively increase the separation between



(a) (b)

Fig. 11. Feature distributions before and after the unknown sample synthesis for the TR dataset. (a) Before synthesis. (b) After synthesis.



(a) (b) (c) (d)

Fig. 12. Sensitivity analysis of parameters on OA. (a) Parameter $\text{Epoch}_{2\text{nd}}$. (b) Parameter $\text{Num}_{\text{proto}}$. (c) Parameter $\text{Dim}_{\text{proto}}$. (d) Parameter $\text{Num}_{\text{pseudo}}$.

known classes and occupy an embedding space distinct from the known categories. This synthesis of unknown classes successfully simulates the presence of unknown classes, thereby validating the rationale behind our method.

E. Parameters and Sensitivity Analysis

In this section, we conducted a sensitivity analysis of the key parameters within the entire framework. These parameters include the number of iterations in the second training stage ($\text{Epoch}_{2\text{nd}}$), the number of multiple Gaussian prototypes ($\text{Num}_{\text{proto}}$), the dimensionality of these prototypes ($\text{Dim}_{\text{proto}}$), and the number of pseudo-unknown classes ($\text{Num}_{\text{pseudo}}$) in the virtual unknown classifier. Each parameter was individually adjusted to assess its impact on OA (%) and the accuracy of the unknown class, as illustrated in Figs. 12 and 13, respectively.

First, the value of $\text{Epoch}_{2\text{nd}}$ indicates how well the model balances the interclass distances among known classes and the discrimination between known and unknown classes. We experimented with various $\text{Epoch}_{2\text{nd}}$ values ranging

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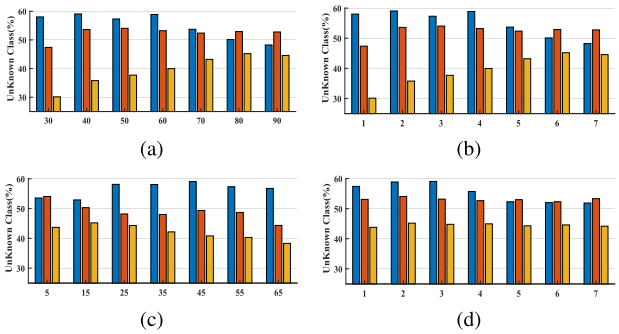


Fig. 13. Sensitivity analysis of parameters on unknown class accuracy. (a) Parameter $\text{Epoch}_{2\text{nd}}$. (b) Parameter $\text{Num}_{\text{proto}}$. (c) Parameter $\text{Dim}_{\text{proto}}$. (d) Parameter $\text{Num}_{\text{pseudo}}$.

from 30 to 90. Figs. 12(a) and 13(a) reveal that the optimal OA and the highest accuracy for unknown classes were achieved at 40, 50, and 80 for the three datasets. As the number of epochs increases, we observe a decline in OA due to a drop in unknown class accuracy. However, within this range, the sensitivity of the $\text{Epoch}_{2\text{nd}}$ parameter is minimal, and small variations do not significantly affect performance.

Second, the number of Gaussian prototypes ($\text{Num}_{\text{proto}}$) is a crucial parameter in the framework, as it influences the network's ability to fit known classes and consequently affects the quality of reconstruction. We tested prototype numbers ranging from 1 to 7, with the results for the three datasets shown in Figs. 12(b) and 13(b). The optimal values are 6, 4, and 1 for the three datasets. Our analysis suggests that the optimal number of prototypes is dataset-dependent. For instance, the TR dataset, which has a more imbalanced category distribution, requires more Gaussian prototypes for effective fitting. At the same time, the MU dataset's simple category distribution requires just one Gaussian prototype per category for fitting.

Third, the dimensionality of the Gaussian prototypes ($\text{Dim}_{\text{proto}}$) affects the complexity of each prototype. We evaluated this parameter across a range of values: {5, 15, 25, 35, 45, 55, 65}. The findings in Figs. 12(c) and 13(c) indicate that this parameter significantly impacts the performance. For the HU dataset, the optimal $\text{Dim}_{\text{proto}}$ is 5, resulting in simpler prototypes. In contrast, the optimal $\text{Dim}_{\text{proto}}$ for the TR and MU datasets is 45 and 15, respectively. This also reflects the complexity of the prototype required to fit the TR dataset.

Finally, the number of unknown categories in the unknown classifier ($\text{Num}_{\text{pseudo}}$) influences the model's ability to identify unknown classes. We tested values from 1 to 7, and Figs. 12(d) and 13(d) demonstrate that the best performance across the three datasets occurs when the number of unknown categories is 3, 2, and 2, respectively. As $\text{Num}_{\text{pseudo}}$ increases, both OA and open-set classification performance tend to decline. This suggests that having more unknown categories does not necessarily improve performance. Instead, it should align with the characteristics of the actual dataset.

F. Computational Complexity Analysis

In this section, we evaluate the computational complexity of the proposed HyLiOSR. We specifically test the training efficiency of the HyLiOSR compared to other related methods.

TABLE XI
EXECUTION TIME (IN SECONDS) OF ONE EPOCH TRAINING OF DIFFERENT METHODS

Dataset	TB-CNN	HRWN	EndNet	GLT-Net	CROSR	MDL4OW	GSRL	HyLiOSR
TR	3.22	8.65	15.39	19.32	9.71	18.73	13.17	16.72
HU	9.81	12.34	20.11	28.31	13.13	22.31	19.71	24.73
MU	6.34	11.23	17.28	25.31	7.33	15.33	11.01	13.22

All experiments were conducted on an NVIDIA GeForce GTX 4090 GPU and the Intel Xeon Silver 4210R CPU. The experiments utilize the open-source deep learning framework PyTorch, Version 2.11. Table XI presents the training time for various algorithms over a single epoch, maintaining consistent parameter settings across the board. Notably, the TB-CNN demonstrates the shortest training time due to its simpler architecture. In contrast, our proposed HyLiOSR exhibits moderate training time while achieving superior results in both known classification and unknown detection tasks.

V. CONCLUSION

In this article, we introduce the innovative HyLiOSR framework, which incorporates reconstruction-based OSR networks and a novel staged progressive training strategy. During the first stage, the reconstruction network focuses on accurately reconstructing known classes by leveraging multiple Gaussian prototypes, thereby ensuring clear differentiation among these classes. In the second stage, we enhance the classifier's ability to identify unknown classes by actively simulating unknown class samples during feature extraction and implementing a virtual unknown classifier. This approach facilitates greater separation between known classes and improves the capacity of the model to distinguish unknown classes. Experimental results across three HSI-LiDAR paired OSR benchmarks indicate that our proposed method surpasses existing advanced techniques. The performance of our algorithm has been greatly improved compared to previous OSR methods, but there is still some room for improvement in recognizing unknown classes. In addition, with the requirement for generalization, future work in the OSR field should try to integrate foundation models to achieve more accurate recognition.

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